Preprocessing Report

Phishing URL Detection Dataset — Week 3 Assignment

INTERN DETAILS

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Week	3 of 8
GitHub	Link

1. Objective

Prepare the raw phishing dataset for machine learning modelling through systematic data cleaning, feature engineering, encoding, normalization, and splitting.

2. Preprocessing Pipeline Summary

Step	Action	Input Shape	Output Shape
1	Load raw data	—	(11430, 89)
2	Remove duplicates	(11430, 89)	(11430, 89)
3	Check missing values	(11430, 89)	(11430, 89)
4	Drop zero-variance features	(11430, 89)	(11430, 83)
5	Drop raw URL string column	(11430, 83)	(11430, 82)
6	Clip negative sentinel values	(11430, 82)	(11430, 82)
7	Label encode target (`status`)	(11430, 82)	(11430, 82)
8	Log1p transform skewed features	(11430, 82)	(11430, 82)
9	RobustScaler on continuous features	(11430, 82)	(11430, 82)
10	Stratified 80/20 train/test split	(11430, 82)	Train: (9144, 82) / Test: (2286, 82)

Final usable feature count: 81 (82 columns minus 1 target column)

3. Step-by-Step Details

3.1 Duplicate Removal

```
Duplicates found: 0
Rows after deduplication: 11,430
```

No duplicate rows were present in the dataset. The deduplication step was confirmed and passed cleanly.

3.2 Missing Value Handling

```
Columns with missing values: 0
Total missing cells        : 0
```

The dataset contains **no missing values** across all 89 original columns. No imputation strategies (mean/median fill, KNN imputation, etc.) were required. This is ideal for downstream modelling without additional data quality concerns.

3.3 Zero-Variance Feature Removal

Six features were identified as **constant** (a single unique value across all 11,430 rows) and dropped:

Feature	Constant Value	Reason for Drop
`nb_or`	0	No `|` characters appear in any URL
`ratio_nullHyperlinks`	0	No null hyperlinks recorded
`ratio_intRedirection`	0	No internal redirections recorded
`ratio_intErrors`	0	No internal errors recorded
`submit_email`	0	No email submission forms detected
`sfh`	0	Server Form Handler always absent

These features contribute **zero discriminative power** and would add unnecessary noise during modelling.

3.4 Raw URL Column Removal

The column `url` (raw string) was dropped. While it could be used for character-level n-gram feature engineering, it is not a structured numerical feature and is not compatible with standard ML pipelines without additional NLP preprocessing.

3.5 Negative Value (Sentinel Code) Handling

Two features contained negative values that are semantically invalid:

Feature	Negative Count	Action
`domain_age`	1,837 rows (16.1%)	Clipped to 0
`domain_registration_length`	46 rows (0.4%)	Clipped to 0

These negatives are **sentinel codes** (e.g., `-1` or `-12`) indicating unknown or unregistered domains — not actual negative time values. Clipping to 0 preserves the "unknown/new" semantics while removing invalid numeric values.

3.6 Feature Encoding

Target variable (`status`):

Original Label	Encoded Value
`legitimate`	0
`phishing`	1

`LabelEncoder` from scikit-learn was applied. All other features were already in numerical form (int64/float64). The 31 binary features (values exclusively 0 or 1) required no additional encoding.

3.7 Feature Type Classification

After dropping removed columns, features were categorized as:

Category	Count	Description
Binary (0/1)	31	Presence/absence flags (e.g., `login_form`, `https_token`, `ip`)
Continuous	50	Count, ratio, and length measurements
Total	**81**	

3.8 Log1p Transformation

42 out of 50 continuous features exhibited absolute skewness > 1 and were transformed using `np.log1p(x)` (i.e., `log(1 + x)`):

- `log1p` is preferred over `log` because it handles zero values gracefully (`log(0)` is undefined).
- Applied only to continuous features; binary features were left unchanged.

Sample skewness reduction:

Feature	Skew Before	Skew After	Reduction
`nb_dollar`	55.66	39.87	−28%
`char_repeat`	15.76	−0.05	−100%
`longest_words_raw`	13.53	1.51	−89%
`avg_word_path`	13.45	−0.43	−103%

Feature	Skew Before	Skew After	Reduction
`nb_extCSS`	23.50	1.85	−92%
`nb_hyperlinks`	7.68	0.67	−91%
`domain_registration_length`	9.82	0.89	−91%
`length_url`	8.09	0.40	−95%

3.9 Feature Scaling — RobustScaler

`RobustScaler` was applied to all 50 continuous features.

Formula:

```
x_scaled = (x - median(x)) / IQR(x)
```

Why RobustScaler over StandardScaler or MinMaxScaler?

Scaler	Sensitive to Outliers?	Best For
StandardScaler	Yes (uses mean/std)	Gaussian-distributed, clean data
MinMaxScaler	Yes (uses min/max)	Bounded data with no outliers
RobustScaler	**No (uses median/IQR)**	**Data with significant outliers ✓**

Given that URL-length and hyperlink count features contain meaningful extreme values (genuine long phishing URLs, pages with thousands of links), `RobustScaler` is the appropriate choice — it scales without being distorted by those extremes.

Post-scaling sample statistics:

Feature	Mean	Std	Min	Median	Max
`length_url`	0.09	0.76	−1.74	0.00	4.71
`length_hostname`	0.04	0.87	−3.11	0.00	5.32
`nb_dots`	0.35	1.03	−1.41	0.00	7.37
`nb_hyphens`	0.62	0.91	0.00	0.00	5.46

Binary features were **not scaled** — their 0/1 encoding is already on a meaningful bounded scale.

3.10 Train / Test Split

An **80/20 stratified split** was applied using `random_state=42` for reproducibility.

Split	Samples	Legitimate	Phishing
Training	9,144 (80%)	4,572 (50%)	4,572 (50%)

Split	Samples	Legitimate	Phishing
Test	2,286 (20%)	1,143 (50%)	1,143 (50%)

Stratification was used to ensure both classes remain perfectly balanced in both splits, preserving the 50/50 distribution from the original dataset.

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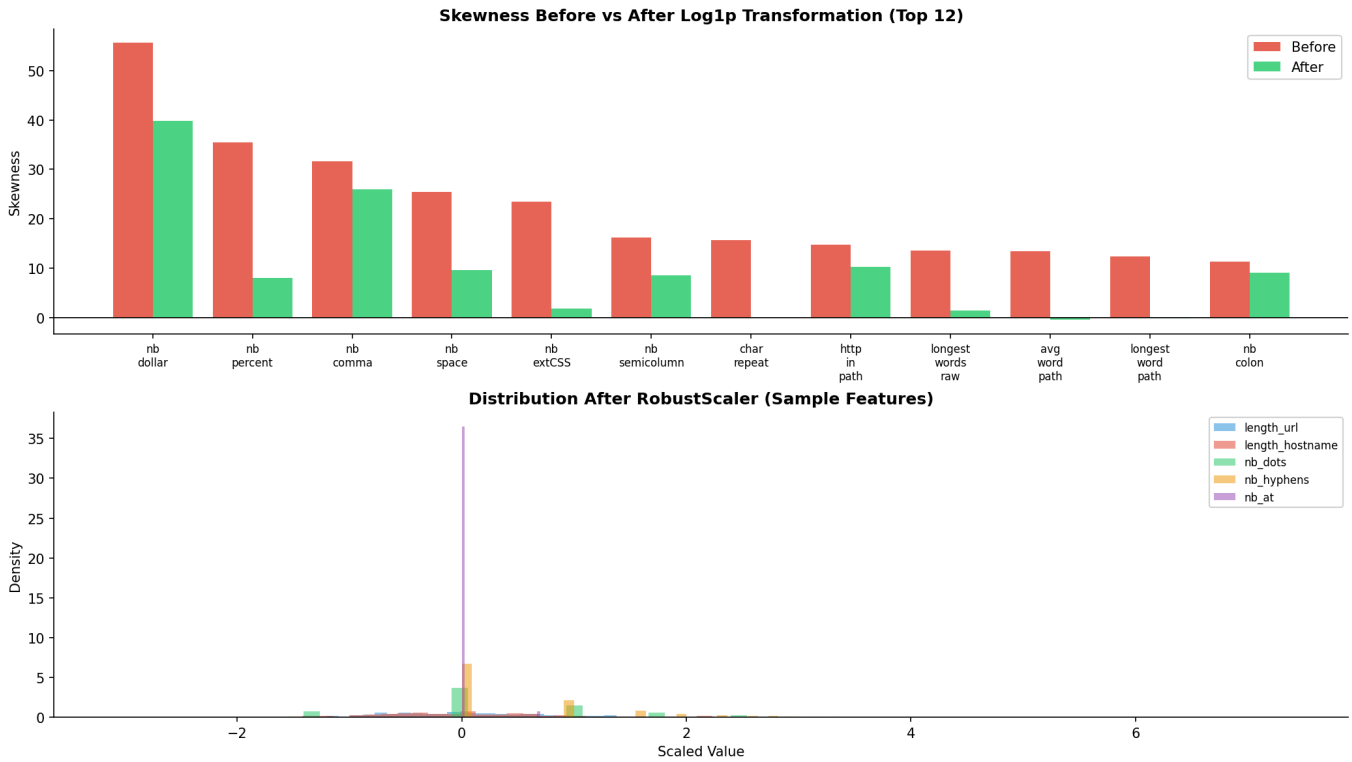
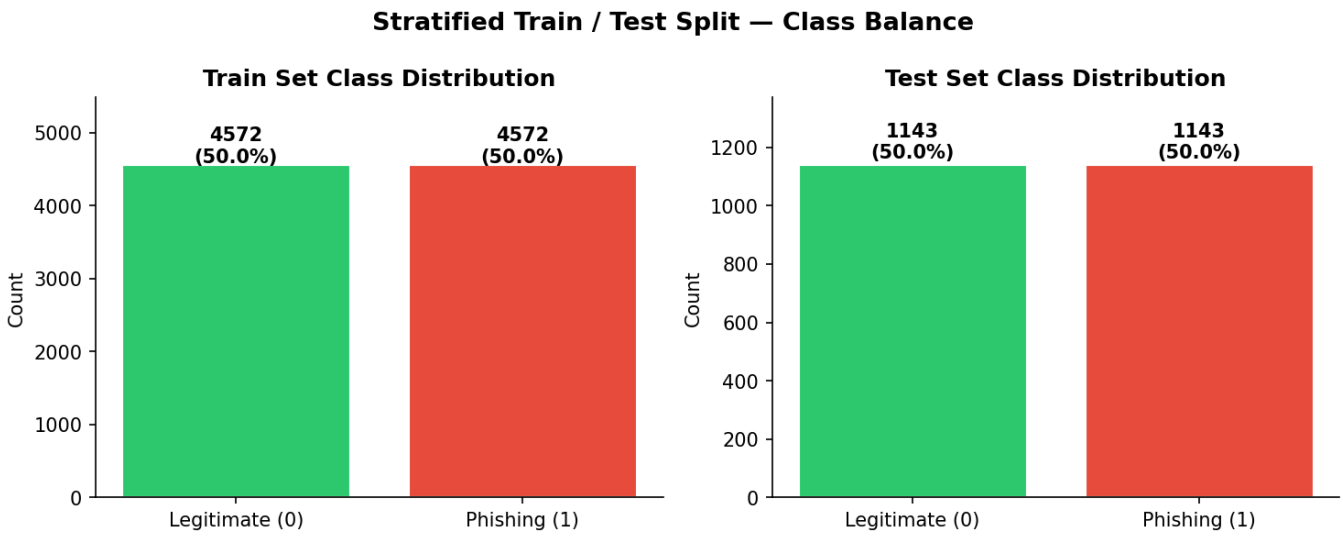
4. Visualizations

Figure	Description
`fig_w3_skew_scale.png`	Skewness before vs after log1p (top 12 features) + scaled distributions
`fig_w3_split_balance.png`	Class balance in train and test splits
`fig_w3_heatmap.png`	Correlation heatmap post-preprocessing (top 20 features by variance)
`fig_w3_variance.png`	Feature variance after scaling across all 81 features

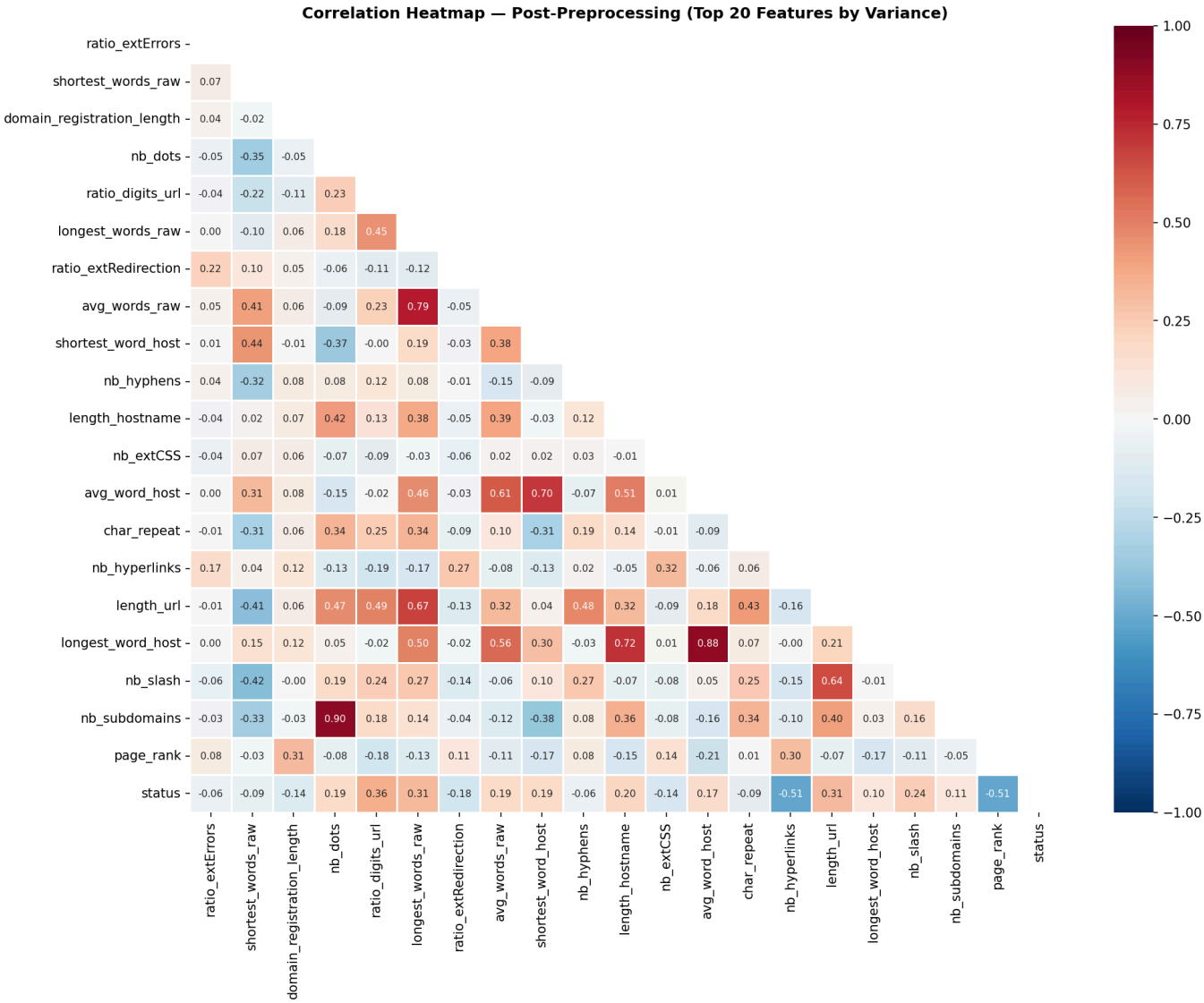
fig_w3_skew_scale.png



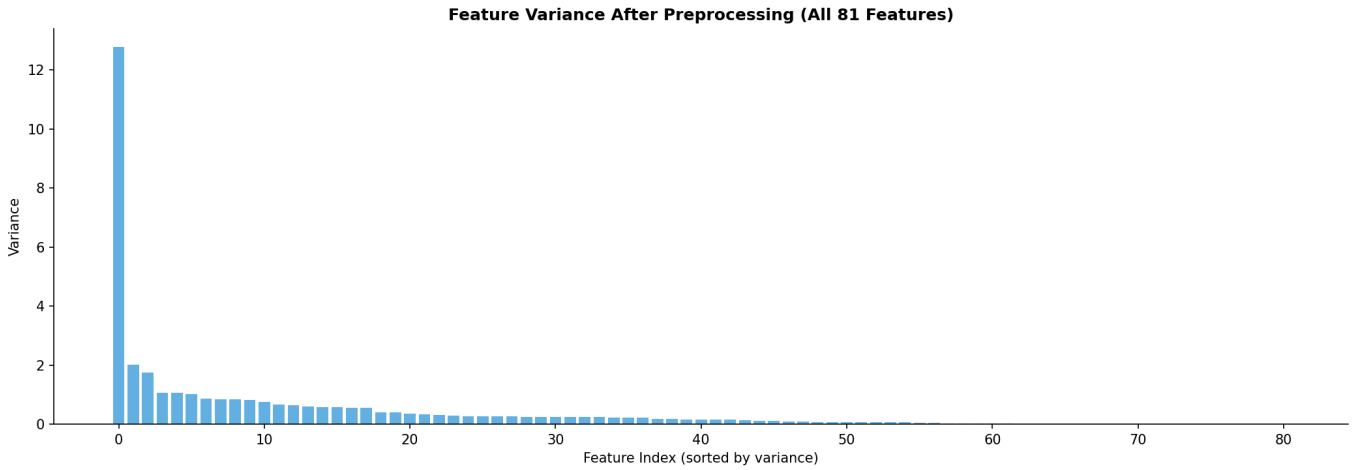
fig_w3_split_balance.png



fig_w3_heatmap.png



fig_w3_variance.png



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5. Output Files

File	Description	Shape
`train_preprocessed.csv`	Training set (features + target)	(9144, 82)
`test_preprocessed.csv`	Test set (features + target)	(2286, 82)
`full_preprocessed.csv`	Full preprocessed dataset	(11430, 82)

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6. Final Dataset Summary

Property	Value
Original features	88
Features after preprocessing	**81**
Features dropped	7 (6 zero-variance + 1 raw URL)
Missing values	0
Duplicates	0
Target encoding	legitimate=0, phishing=1
Continuous features scaled	50 (RobustScaler)
Skewed features transformed	42 (log1p)
Binary features	31 (unchanged)
Train size	9,144
Test size	2,286
Class balance (both splits)	50% / 50% (perfectly stratified)

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7. Recommendations for Week 4 (Modelling)

- The preprocessed `train_preprocessed.csv` and `test_preprocessed.csv` files are **model-ready** with no further cleaning required.
- For tree-based models (Random Forest, XGBoost, LightGBM): scaling is not strictly necessary, but the log1p transformation still benefits by compressing extreme values.
- For linear models (Logistic Regression, SVM, Neural Networks): the RobustScaler output is critical for stable gradient behaviour.
- Consider **feature selection** (e.g., Recursive Feature Elimination or feature importance from a quick Random Forest) before training, as 81 features may include redundant ones (several binary flags had near-zero class separation in Week 2 EDA).
- The `page_rank`, `domain_age`, `ratio_digits_url`, and `nb_hyperlinks` features are expected to be the top contributors based on correlation analysis from Week 2.

Report prepared for Week 3 Preprocessing Assignment — Phishing URL Detection
Dataset: 11,430 rows | Pipeline: dedup → missing check → feature drop → encoding → log1p → RobustScaler → stratified split